

1. What is a Data Structure? Why do we need data structures?

Follow-up Questions:

- Can you give some real-life examples?
- Which data structure is used to store student records?

2. What is the difference between Linear and Non-Linear Data Structures?

Follow-up Questions:

- Give two examples of each.
- Which one is easier to traverse?

3. Explain the difference between an Array and a Linked List.

Follow-up Questions:

- Which one provides faster random access?
- Which one is better for frequent insertions and deletions?
- Why?

4. What is Time Complexity?

Follow-up Questions:

- What does Big-O notation represent?
- Why is time complexity important?
- Which is better: $O(n)$ or $O(\log n)$?

5. Explain the following time complexities with one example each.

- $O(1)$
- $O(\log n)$
- $O(n)$
- $O(n \log n)$
- $O(n^2)$

Follow-up Questions:

- Which is the most efficient?
- Which sorting algorithm has $O(n \log n)$ average complexity?

6. What is the difference between a Stack and a Queue?

Follow-up Questions:

- What are their basic operations?
- Give one real-life application of each.
- Which principle does each follow?

7. What is Recursion?

Follow-up Questions:

- What is the base case?
- What happens if the base case is missing?
- Can every recursive solution be converted into an iterative one?

8. What is the difference between Linear Search and Binary Search?

Follow-up Questions:

- Which one is faster?
- What condition is required before applying Binary Search?
- What are their time complexities?

9. Explain Bubble Sort and Selection Sort.

Follow-up Questions:

- Which performs fewer swaps?
- What is the worst-case time complexity of both?
- Which one would you prefer and why?

10. What is a Binary Tree?

Follow-up Questions:

- What is the difference between a Binary Tree and a Binary Search Tree (BST)?
- What are Inorder, Preorder, and Postorder traversals?
- Which traversal returns sorted data in a BST?